

Figure 1: Overview of the AKT method. We use the Rasch model-based embeddings as raw embeddings for questions and responses. The question and knowledge encoders compute the context-aware representations of questions and responses pairs. The knowledge retriever uses these representations as input and computes the knowledge state of the learner. For simplicity, we do not show the monotonic attention mechanism in the encoders. We do not show sublayers either.

with

$$s_{t;\tau} = \frac{\exp(-\theta \cdot d(t, \tau)) \cdot \mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{D_k}}, \quad (1)$$

where $\theta > 0$ is a learnable decay rate parameter and $d(t, \tau)$ is temporal distance measure between time steps t and τ . In other words, the attention weights for the current question on a past question depends not only on the similarity between the corresponding query and key, but also on the relative number of time steps between them. In summary, our monotonic attention mechanism takes the basic form of an exponential decay curve over time with possible spikes at time steps when the past question is highly similar to the current question. We note that we apply exponential decay to the attention weights rather than latent knowledge, which is the common approach in existing learner models (see e.g., [17, 26]).

We note that there are many other possible ways to characterize the temporal dynamics of attention. First, in language tasks that attention networks excel at, the temporal dynamics can be modeled using additive positional embeddings or learnable embeddings [29]. Second, in our monotonic attention mechanism, we can also parameterize the exponential decay as $s_{t;\tau} = \frac{\mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{D_k}} - \theta \cdot d(t, \tau)$. However, neither of these changes lead to comparable performance to our chosen model setup; we will compare AKT against its variants using positional encoding rather than monotonic attention in our experiments.

A context-aware distance measure. The exponential decay function decides the rate at which the attention weights decay as the distance between the current time index and the previous time indices increase. A straightforward way to define the distance between two time indices is their absolute value difference, i.e.,

$d(t, \tau) = |t - \tau|$. However, this distance is not context-aware and ignores the practice history of each learner. For example, consider the two following sequences of concepts a learner practiced on:

Venn Diagram (VD)₁, VD₂, ..., VD₈, Prime Numbers (PN)₉, PN₁₀
and

$$\text{PN}_1, \text{VD}_2, \text{VD}_3, \dots, \text{VD}_9, \text{PN}_{10},$$

where the notation “VD₂” means that the learner practiced the concept of Venn Diagram at time step 2. In this example, the learner answers a question on Prime Numbers at $t = 10$, i.e., the current time index, in both of these sequences, but the most recent past practice on Prime Numbers comes at different time indices. Since the concepts of Venn Diagram and Prime Numbers are not closely related, the learner’s previous practices on Prime Numbers is more relevant to us when predicting their answer to the current practice question than recent practices on Venn Diagram. In this case, with the straightforward absolute value difference, an exponential decay curve will significantly reduce the attention weight assigned to the practice on Prime Numbers at $t = 1$.

Therefore, we propose the following context-aware distance measure between time steps $d(t, \tau)$ with $\tau \leq t$ for the exponential decay mechanism (in the encoders):

$$d(t, \tau) = |t - \tau| \cdot \sum_{t'=\tau+1}^t \gamma_{t;t'},$$

$$\gamma_{t;t'} = \frac{\exp(\frac{\mathbf{q}_t^\top \mathbf{k}_{t'}}{\sqrt{D_k}})}{\sum_{1 \leq \tau' \leq t} \exp(\frac{\mathbf{q}_t^\top \mathbf{k}_{\tau'}}{\sqrt{D_k}})}, \quad \forall t' \leq t.$$

“Conversion of Fractions to Percents”, and their learned difficulty parameters. We show three questions for each concept: an easy one, an average one, and a hard one. Using the “Probability of a Single Event” concept as an example, the learned difficulty parameter values (μ_q) are -0.0515 for the easy one, 0.0088 for the average one, and 0.0548 for the hard one. These learned difficulty levels match our understanding about the difficulty levels of these questions.

These results suggest that AKT has the potential to be applied in real-world educational settings. Using the estimated difficulty parameters, a computerized learning platform can either i) automatically select questions with appropriate difficulty levels for each learner given their past responses, or ii) support teachers to adjust course plans by providing them feedback on question difficulty levels learned from real data. Therefore, AKT improves over existing KT methods by not only providing state-of-the-art predictive performance but also exhibiting interpretability and potential for personalized learning.

5 CONCLUSIONS AND FUTURE WORK

In this paper, we have proposed attentive knowledge tracing, a new method for knowledge tracing that relies fully on attention networks. Our method improves upon existing knowledge tracing methods by building context-aware representations of questions and responses, using a monotonic attention mechanism to summarize past learner performance in the right time scale, and by using the Rasch model to capture individual differences among questions covering the same concept. Experimental results on a series of benchmark real-world learner response datasets show that our method outperforms state-of-the-art KT methods and exhibit excellent interpretability. Avenues of future work include i) incorporating question text to further enhance the interpretability of question and concept embeddings and ii) testing whether our method can improve prediction performance on language learning datasets where memory decay occurs [26].

REFERENCES

- [1] Hao Cen, Kenneth Koedinger, and Brian Junker. 2006. Learning factors analysis—A general method for cognitive model evaluation and improvement. In *Proc. International Conference on Intelligent Tutoring Systems*. 164–175.
- [2]

Supplementary Material

Parameter tuning. For all methods, we use a two-layer, fully-connected network for the response prediction model, with 512 and 256 hidden units in the first and second layers, respectively. For AKT, we fix the number of attention heads to $H = 8$ and use the same dimension for the queries, keys, and values for each head in the encoders and the knowledge retriever, i.e., $D_K = D_V = D/8$ where D is the input embedding dimension. For AKT, we share the query and key embedding layer in the attention mechanism. We do not perform grid search over the values of these parameters since the performance of KT methods is insensitive to these

parameters. For all methods, we use $\{256, 512\}$, $\{256, 512\}$, and $\{0, 0.05, 0.1, 0.15, 0.2, 0.25\}$ as values of the input embedding dimension, the hidden state dimension, and the dropout rate for the feedforward network, respectively. We also use $\{1, 10, \infty\}$ as values of the maximum gradient norm for clipping [20] where ∞ means no gradient norm clipping. For AKT, SAKT, and their variants, we use $\{5 \times 10^{-6}, 10^{-5}, 10^{-4}\}$ as values of the learning rate in the Adam optimizer. For DKT, DKT+, and DKVMN, we use $\{10^{-4}, 10^{-3}\}$ as val-